Whether the Sketch Style Itself Produced the Gain

Working Title

Whether the Sketch Style Itself Produced the Gain.

Problem Statement

Raising SketchThinker-R1's style-reward weight from 0.05 to 0.1 moves tokens only from 65.2 to 64.3, and the cold-start rewrite is produced by GPT-5, which can change content as well as length. The reported 64% reduction and the accuracy are therefore not yet attributed to the sketch style.

Proposed Solution

This plan does not propose a new judge. Four arms share the base model, the items, the decoding, and three seeds: the paper's sketch pipeline; a GPT-5 rewrite that keeps the original length and style, then the same training; reinforcement learning with the style reward set to zero; and answering with no reasoning. The horizontal axis is end-to-end latency, including image tokens, not accuracy divided by reasoning tokens. The style can be credited only if the sketch arm beats both the length-preserving rewrite and the zero-style arm. A win over the length-preserving rewrite alone would still be compatible with the rewriter having changed the content.

Key Research Questions

- Once the rewriter is held fixed, does shortening still beat a length-preserving rewrite?
- Does setting the style reward to zero change length or accuracy relative to the default weight?
- On items where image tokens dominate latency, does cutting reasoning tokens still move end-to-end latency?

Expected Outcome

The study will attribute the reported gain to the sketch style, to the rewriter, to reinforcement learning, or to the metric.

Comparison

Arm	What it holds fixed	What a sketch loss would mean
Length-preserving rewrite	Same rewriter, original length	The gain came from changing content
Style reward at zero	Same cold start, no style reward	The style reward did not cause the shortening
Direct answer	No reasoning trace	The trace was not carrying the accuracy